

SCS2315 – Electronics & Physical Computing

Dr. H.E.M.H.B. Ekanayake

Course Information

Course Coordinator / Lecturer: Dr. Hiran Ekanayake (HBE)

Teaching Assistants: Mr. Janitha Dissanayake (JSB) & others

Duration: 15 Weeks, starting from **November 3rd**

Lecture Schedule:

Mondays: 1:00 – 3:00 PM (Room S104)

Practicals / Tutorials: Wednesdays & Fridays (you will be assigned to groups)

Students **must bring their laptops** for practical sessions with the **required software tools (e.g., Arduino IDE)** pre-installed.

Course Descriptor

Learning Outcomes

By the end of this course, students will be able to:

- **Explain** fundamental concepts of voltage, current, resistance, power, and circuit laws.
- **Differentiate** between passive, active, and integrated components, and **analyze** their roles in circuits.
- **Apply** Boolean algebra, digital logic, and circuit principles to implement combinational and sequential systems.
- **Program** microcontrollers (Arduino, ESP32) to **control** sensors, actuators, and communication interfaces.
- **Integrate** wired and wireless communication protocols (UART, I²C, SPI, Wi-Fi, BLE, MQTT) into embedded applications.
- **Use** laboratory instruments and EDA/simulation tools to **test** and **evaluate** circuit performance.
- **Design** and **prototype** physical computing systems that combine hardware and software for real-world applications.
- **Assess** safety requirements and **demonstrate** best practices in electronics experimentation and prototyping.

Course Outline

1. Electronic Fundamentals

- Voltage, Current, Resistance, Power, Impedance
- Ohm's Law, KVL/KCL, Thevenin & Norton Theorems

Assessment

Continuous Assessment – 40%

(Average of the best 2 in class assignment marks * 60% + Practical test/s marks * 40%) * Attendance to lectures %

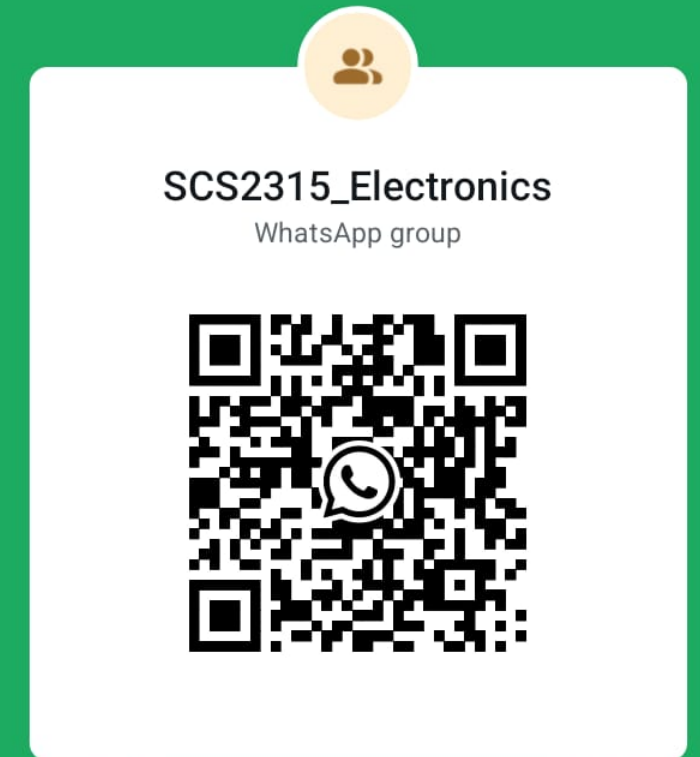
E.g., in class 1 = 80%, in class 2 = 60%, in class 3 = 40%, practical test = 50%, number of lectures conducted = 15, number of lectures attended = 10 → Final assignment marks = $((80+60)/2 * 0.6 + 50 * 0.4) * (10/(15-2)) = (70 * 0.6 + 50 * 0.4) * 0.77 = 62 * 0.77 = 48$

Final Written Paper – 60%

30 MCQs and one structured question

Note: Answers to MCQs need justified giving calculations etc.

Join the WhatsApp Group



Scan this QR code using the WhatsApp camera to join this group

Attendance and Conduct

Entry to the lecture **will not be permitted after 1:15 PM**, and students **must remain until the end** of the session.

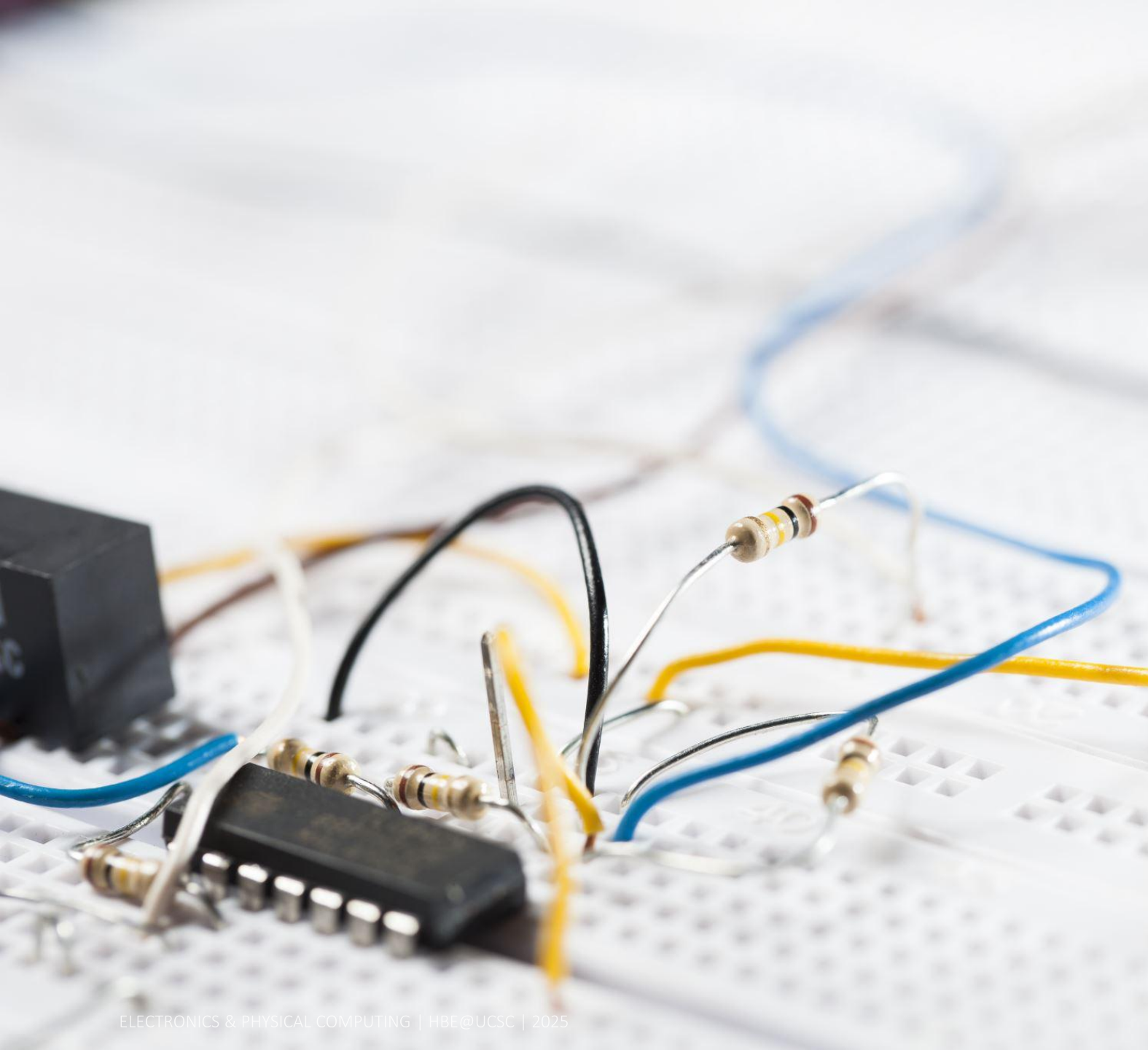
You are expected to **focus solely on the lecture** and avoid engaging in other tasks during class time.

From time to time, short exercises will be given - please bring a **notebook or writing book** for this purpose.

Failure to follow these instructions may result in being asked to leave the lecture and could lead to further disciplinary action.

Lesson 1-1: Foundations of Electronics and Signals

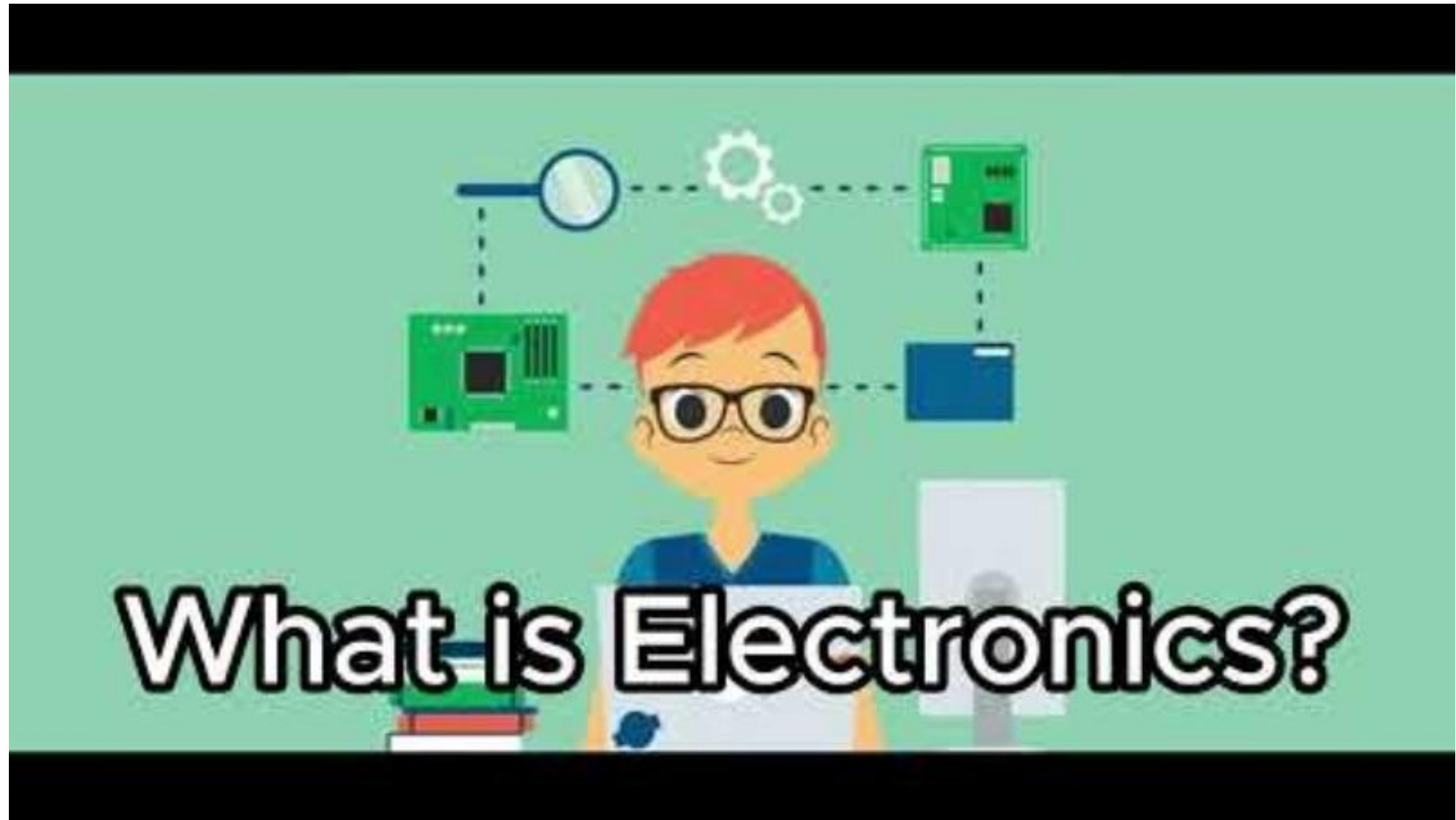
Electronics & Physical Computing



What is Electronics?

Electronics is the branch of science and technology that deals with the study, design, and use of devices and systems that operate by controlling the flow of **electrons** or other charge **carriers** (like holes in semiconductors).

In simpler terms:
Electronics is about using **electric current** to process, transmit, or store information.



<https://www.youtube.com/watch?v=OwLecLDXcyo>

Key Concepts in Electronics

Electric Current – The flow of electric charge, usually through conductors like wires.

Circuits – Paths through which electric current flows. Circuits can be simple (like a battery and a bulb) or complex (like inside a smartphone).

Components – Basic building blocks like:

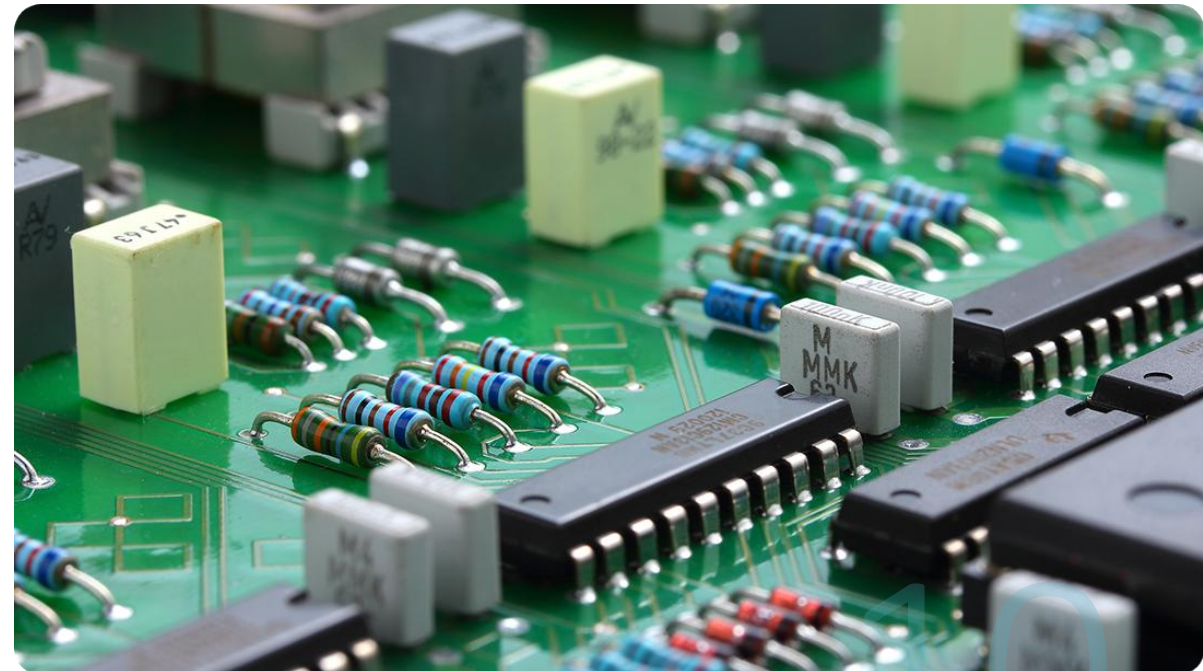
- Resistors (control current)

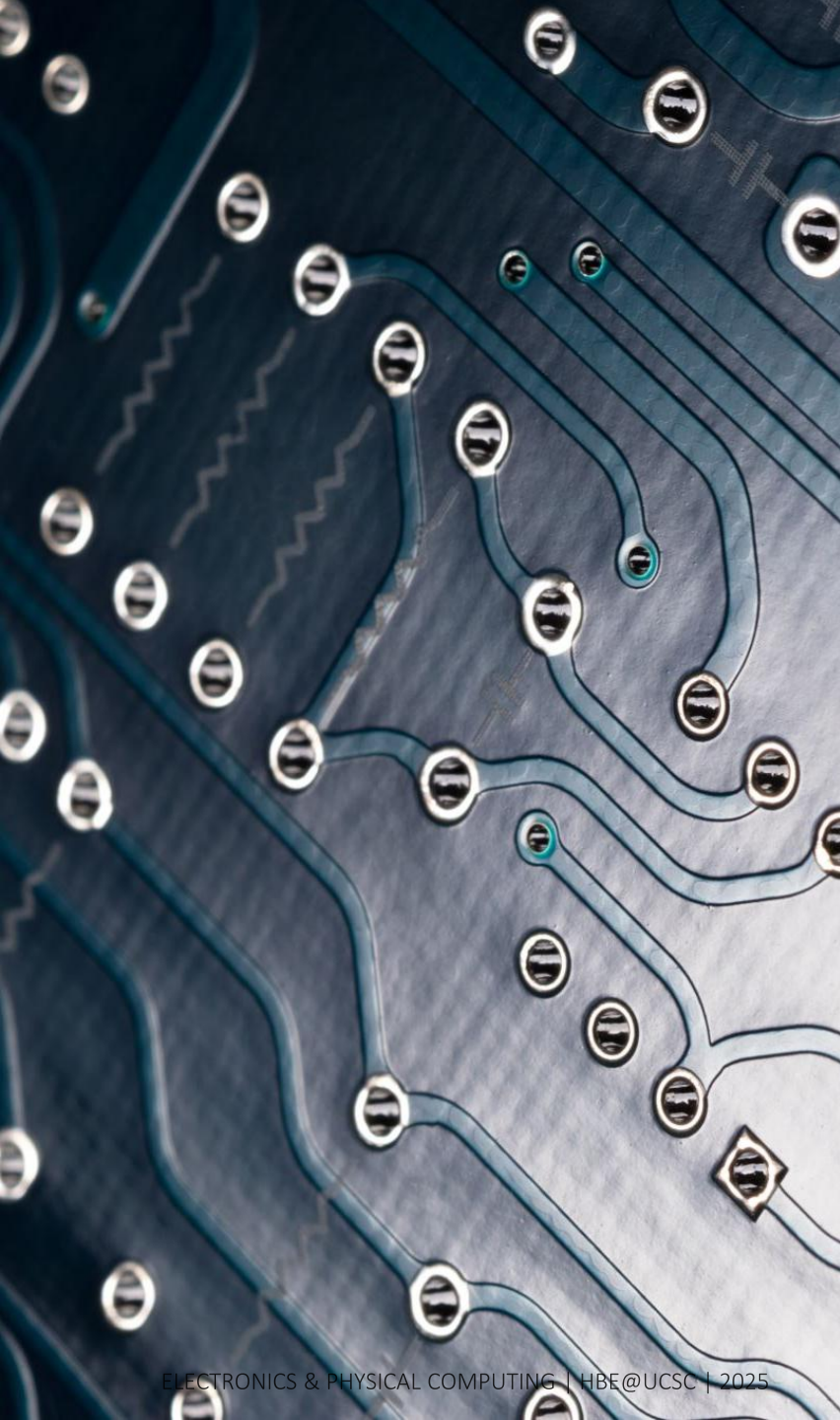
- Capacitors (store energy)

- Inductors (store energy in magnetic fields)

- Diodes (let current flow in one direction)

- Transistors (act as switches or amplifiers)



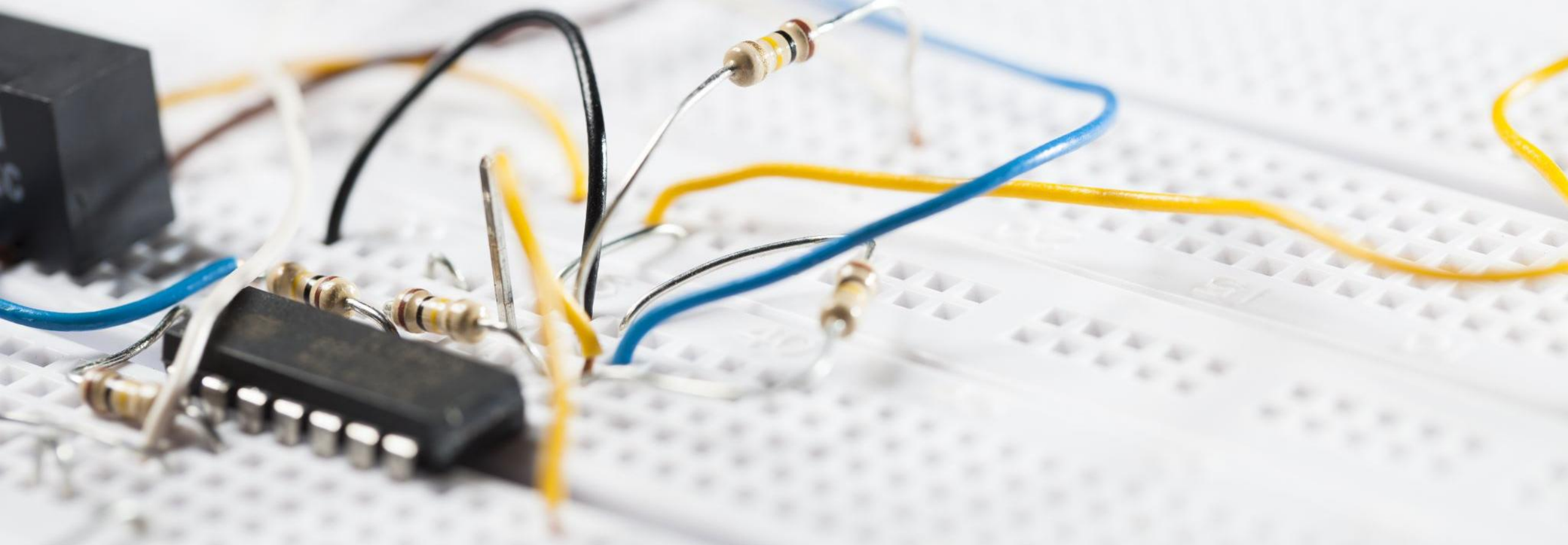


Key Concepts in Electronics

Analog vs. Digital Electronics:

Analog deals with continuous signals (e.g., audio signals).

Digital deals with discrete signals (e.g., 0s and 1s in computers).



Electrical Quantities

What is Electricity?



Voltage



Current

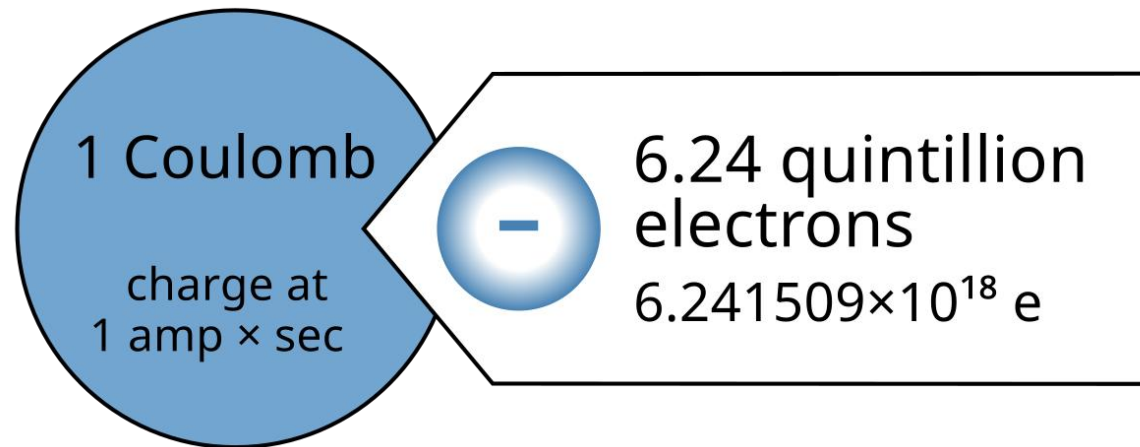


Resistance

<https://www.youtube.com/watch?v=AkWyNFPpRE0>

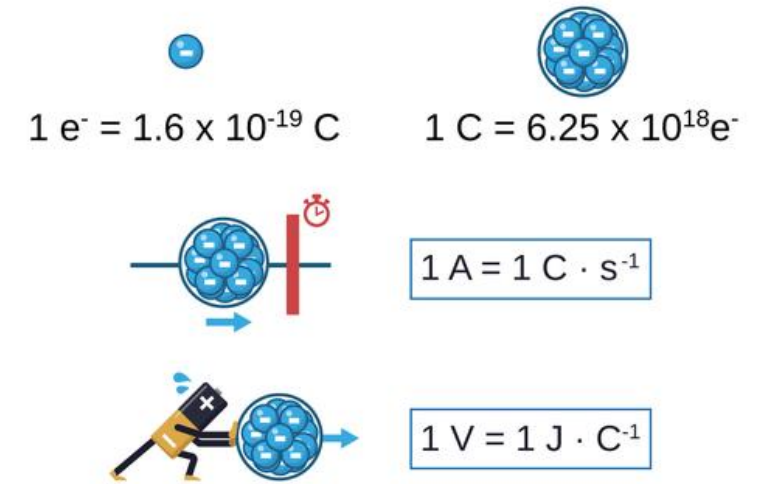
1. Charge (Q)

- **Definition:** The fundamental property of matter responsible for electric phenomena.
- **Unit: Coulomb (C)**
- **Example:** A single electron has a charge of about $-1.6 \times 10^{-19} \text{ C}$.



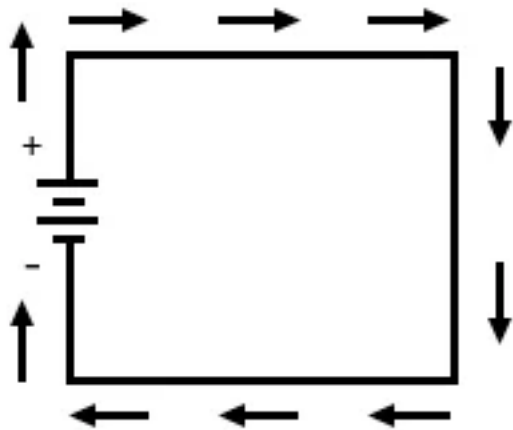
2. Current (I)

- **Definition:** The rate of flow of electric charge.
- **Formula:**
$$I = \frac{Q}{t}$$
where Q is charge, t is time.
- **Unit:** Ampere (A)
- **Example:** If 1 coulomb of charge flows in 1 second, that's 1 ampere of current.



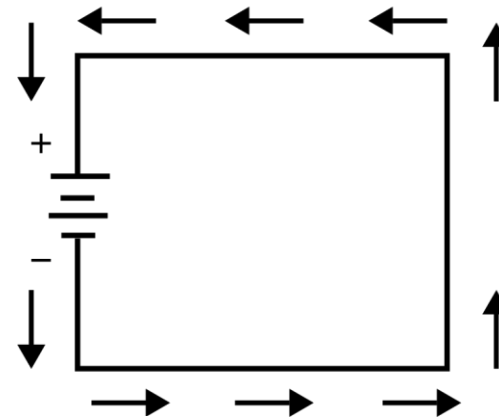
Note

Conventional Flow Notation

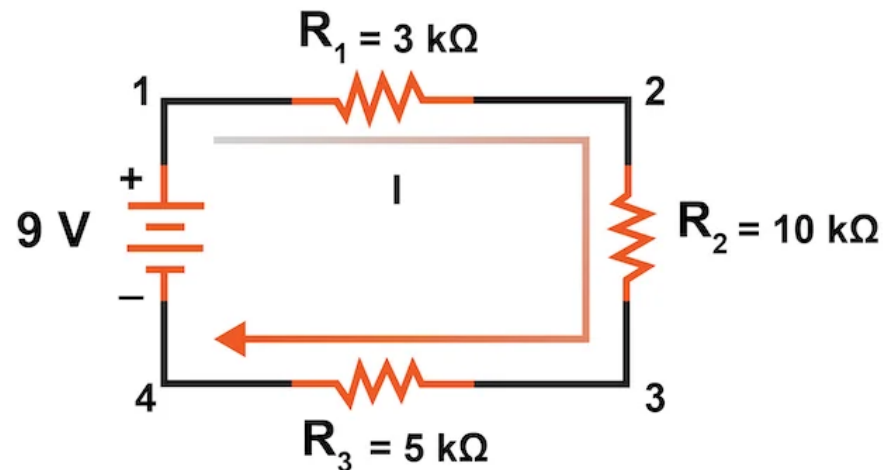


Electric charge moves from the positive (surplus) side of the battery to the negative (deficiency) side.

Electron Flow Notation



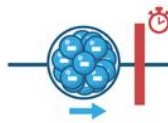
Electron charge moves from the negative (surplus) side of the battery to the positive (deficiency) side.



3. Voltage (V)

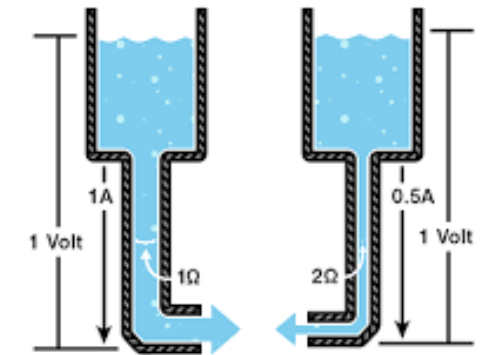
 $1 e^- = 1.6 \times 10^{-19} C$

 $1 C = 6.25 \times 10^{18} e^-$

 $1 A = 1 C \cdot s^{-1}$

 $1 V = 1 J \cdot C^{-1}$

- **Definition:** The **potential difference** between two points; it's the “push” that drives current.
- **Formula:**
$$V = \frac{W}{Q}$$
where W is work or energy in joules, Q is charge.
- **Unit:** Volt (V)
- **Example:** A 9V battery can push charges through a circuit with 9 volts of potential difference.



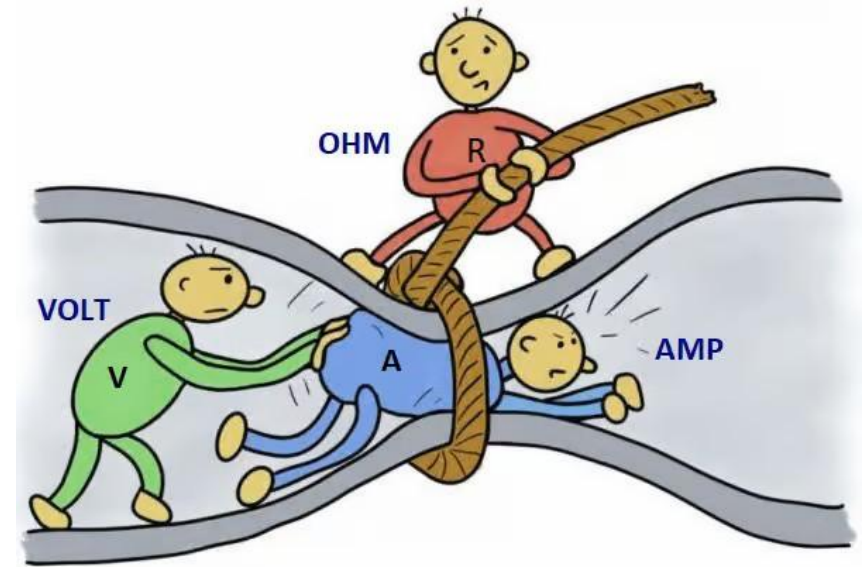
4. Resistance (R)

- **Definition:** The **opposition to the flow of current** in a material.

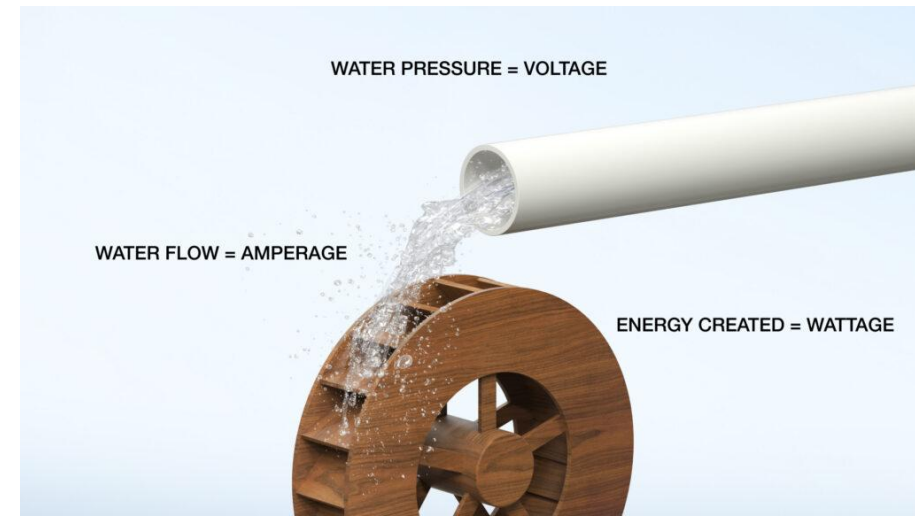
- **Formula:**

$$R = \frac{V}{I}$$

- **Unit:** Ohm (Ω)
- **Example:** A resistor that allows 1 amp of current to flow when 5 volts are applied has a resistance of 5 ohms.



5. Power (P)



- **Definition:** The rate at which electrical energy is consumed or generated.
- **Formula:**
$$P = VI$$

(Also: $P = I^2 R$ or $P = \frac{V^2}{R}$)
- **Unit:** Watt (W)
- **Example:** A 60W bulb consumes 60 joules of energy every second.

6. Energy (E)

- **Definition:** The **capacity to do work**; in electronics, it's how much power is used over time.
- **Formula:**
$$E = Pt$$
- **Unit:** Joule (J) or Watt-hour (Wh)
- **Example:** A 100W appliance running for 2 hours uses 200 Wh of energy.



7. Reactance (X)

- **Definition:** The **opposition to AC current** caused by **inductors and capacitors** (not resistors).

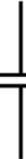
- **Types:**

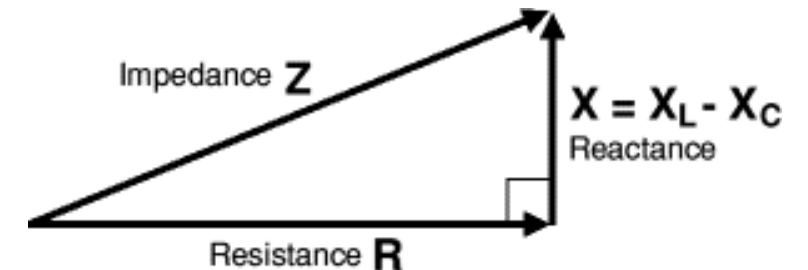
- **Inductive Reactance (X_L):** $X_L = 2\pi fL$

- **Capacitive Reactance (X_C):** $X_C = \frac{1}{2\pi fC}$

- **Unit:** Ohm (Ω)

- **Note:** Reactance depends on **frequency (f)** of the signal.

Resistor 100 Ω  $R = 100 \Omega$ $X = 0 \Omega$ $Z = 100 \Omega \angle 0^\circ$	Inductor 100 mH 159.15 Hz  $R = 0 \Omega$ $X = 100 \Omega$ $Z = 100 \Omega \angle 90^\circ$	Capacitor 10 μF 159.15 Hz  $R = 0 \Omega$ $X = 100 \Omega$ $Z = 100 \Omega \angle -90^\circ$
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$$\text{Impedance, } Z = \sqrt{R^2 + X^2}$$

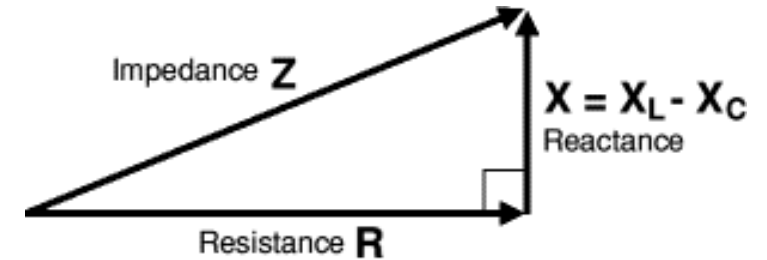
8. Impedance (Z)

- **Definition:** The total opposition to AC that includes both resistance and reactance.

- **Formula:**

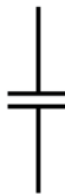
$$Z = R + jX$$

(where j is the imaginary unit, indicating phase difference)



$$\text{Impedance, } Z = \sqrt{R^2 + X^2}$$

- **Unit:** Ohm (Ω)
- **Example:** In AC circuits, impedance determines how signals behave through components.

Resistor 100 Ω	Inductor 100 mH 159.15 Hz	Capacitor 10 μ F 159.15 Hz
		
$R = 100 \Omega$ $X = 0 \Omega$ $Z = 100 \Omega \angle 0^\circ$	$R = 0 \Omega$ $X = 100 \Omega$ $Z = 100 \Omega \angle 90^\circ$	$R = 0 \Omega$ $X = 100 \Omega$ $Z = 100 \Omega \angle -90^\circ$

Electrical Quantity	Water Analogy	Meaning / Visualization
Charge (Q)	Amount of water	Total "stuff" flowing
Current (I)	Flow rate of water (liters/sec)	How fast the water flows
Voltage (V)	Water pressure	The "push" behind the flow
Resistance (R)	Narrow pipe	Harder to flow through
Power (P)	Water force $\times$ flow rate	Work being done (e.g., spinning a waterwheel)
Energy (E)	Total water delivered over time	How much total work is done
Reactance (X)	Flexible or elastic pipe	Temporarily resists changes in flow (frequency dependent)
Impedance (Z)	Combined narrow and elastic pipe	Total opposition to flow in AC circuits

Electrical Water Analogy

Exercise

Q1. A current of **2 A** flows through a conductor for **3 minutes**. Find the **total charge** that passes through the conductor.

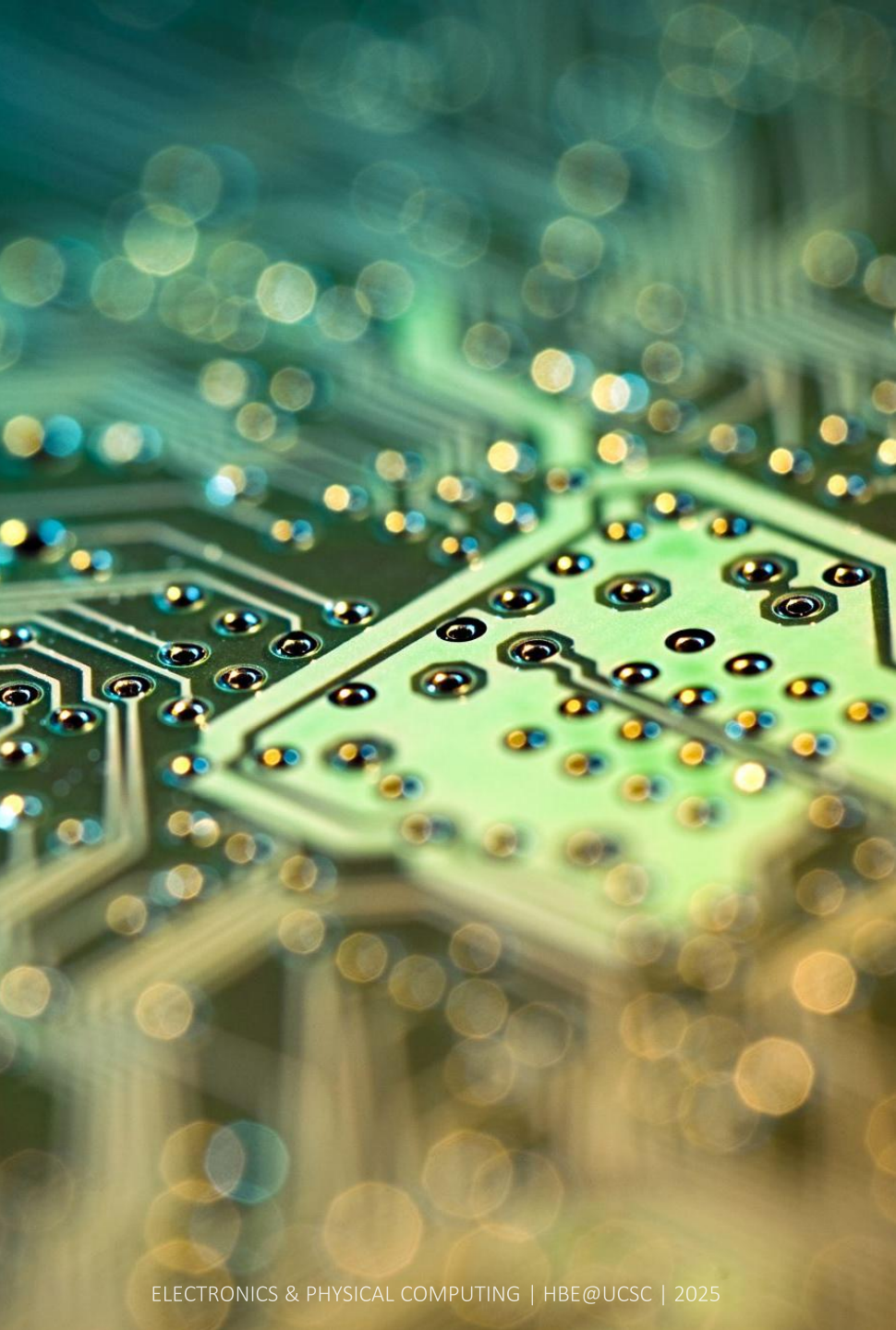
Q2. How many **electrons** flow through a wire when a **charge of 0.5 C** passes? (Charge of one electron = $1.6 \times 10^{-19} \text{ C}$)

$$Q = I \times t$$

$$Q = 2 \text{ A} \times (3 \times 60) \text{ s} = 2 \times 180 = 360 \text{ C}$$

$$n = \frac{Q}{e}$$

$$n = \frac{0.5}{1.6 \times 10^{-19}} = 3.125 \times 10^{18}$$



Basic Circuit Laws

Ohm's Law

$$V = I \times R$$

Ohm's Law defines the relationship between **voltage (V)**, **current (I)**, and **resistance (R)** in an electrical circuit.

V (Voltage): Potential difference across the resistor (volts, V)

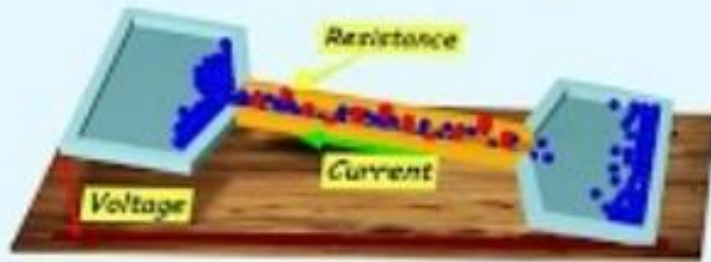
I (Current): Flow of electric charge (amperes, A)

R (Resistance): Opposition to current flow (ohms, Ω)

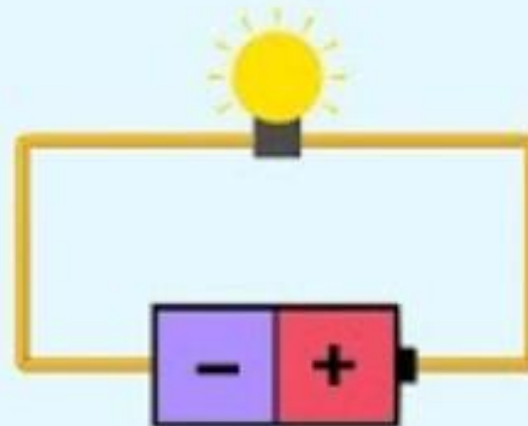
Note: The symbols “E” and “V” are interchangeable for the most part, although some texts reserve “E” (stands for “Electromotive force”) to represent voltage across a source (such as a battery or generator) and “V” to represent voltage across anything else.

Ohm's Law

(Voltage, Current, Resistance)



$$I = \frac{V}{R} \quad V = IR \quad R = \frac{V}{I}$$



Ohm's Law

$$R = \frac{V}{I}$$
$$= \frac{220}{5} \Omega$$
$$= 44 \Omega //$$

<https://www.youtube.com/watch?v=HpEhlnh-hAE>

Exercise 1 (Ohm's Law)

A 12 V battery is connected across a resistor of 8 Ω . Find the current flowing through the resistor.

$$V = I \times R$$

$$I = \frac{V}{R} = \frac{12}{8} = 1.5 \text{ A}$$

Exercise 2 (Ohm's Law)

A current of 2.5 A flows through a resistor when a voltage of 20 V is applied across it. Find the resistance of the resistor.

$$V = I \times R \Rightarrow R = \frac{V}{I}$$

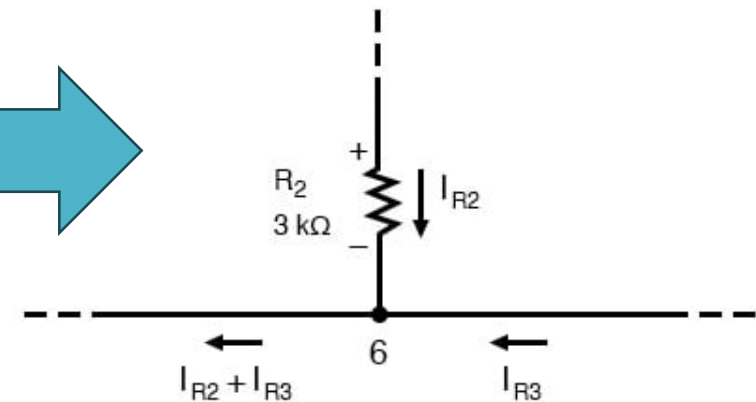
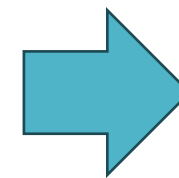
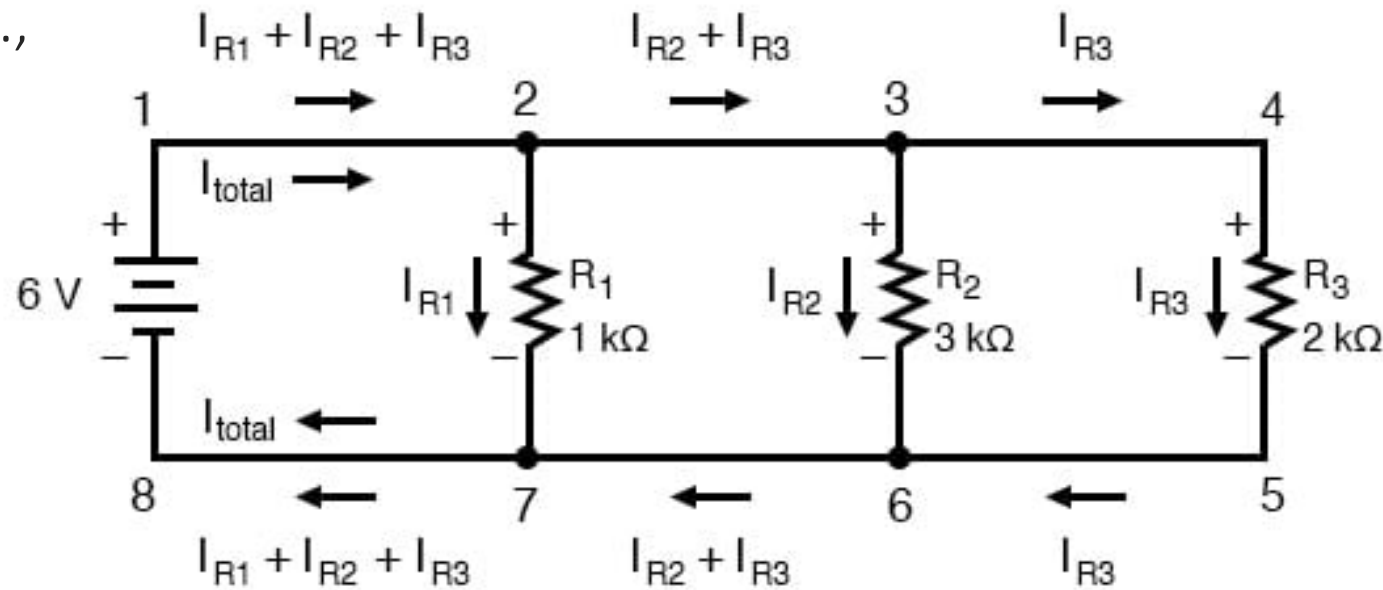
$$R = \frac{20}{2.5} = 8\ \Omega$$

Kirchhoff's Current Law (KCL)

$$\sum I_{\text{in}} = \sum I_{\text{out}}$$

At any junction (node) in a circuit, the **sum of currents entering the node equals the sum of currents leaving the node**, or in other words, the **algebraic sum of all currents entering and exiting a node must equal zero**.

E.g.,



Exercise 1 (KCL)

Problem: A node has a current source injecting 6.0 A into it. The node is connected to ground through $R_1 = 2.0 \text{ } \Omega$ and $R_2 = 3.0 \text{ } \Omega$. Find the node voltage V .

KCL: currents leaving node through resistors = injected current

$$\frac{V}{R_1} + \frac{V}{R_2} = 6.0$$

Answer: $V = 7.2 \text{ V}$.

Exercise 2 (KCL)

Problem: Two nodes A and B (relative to ground) are connected as follows:

- Node A to ground through $R_A = 2 \Omega$.
- Node B to ground through $R_B = 6 \Omega$.
- Node A connected to Node B by $R_{AB} = 4 \Omega$.
- Current source $I_1 = 3.0 \text{ A}$ injects into Node A.
- Current source $I_2 = 1.0 \text{ A}$ injects into Node B.

Find V_A and V_B .

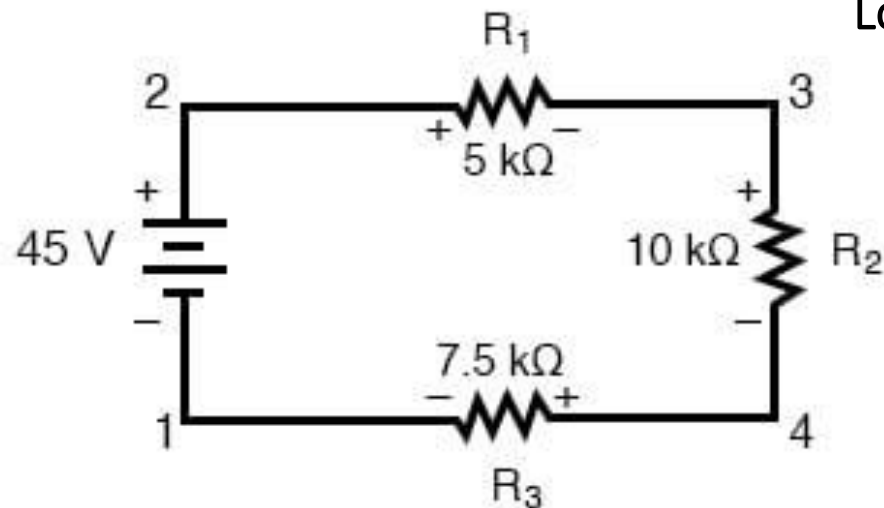
$$3 = \frac{V_A}{2} + \frac{V_A - V_B}{4}, \quad 1 = \frac{V_B}{6} + \frac{V_B - V_A}{4}. \quad \text{Answer: } V_A = 6.0 \text{ V}, \quad V_B = 6.0 \text{ V}.$$

$$\sum V = 0$$

Kirchhoff's Voltage Law (KVL)

In any closed loop in a circuit, the **sum of all voltages** (rises and drops) equals zero, or in other words, the **algebraic sum of all voltages** in a loop must equal zero.

E.g.,



Loop Order: 1-2-3-4-1

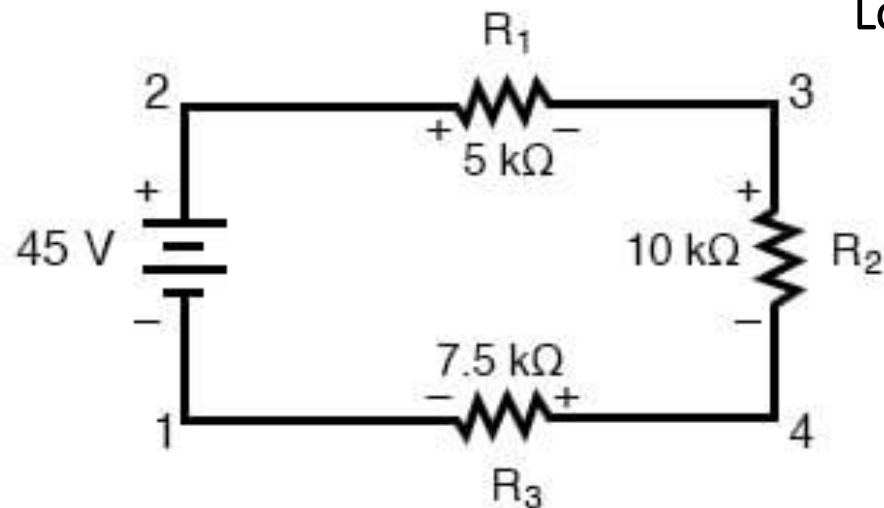
$E_{2-1} = +45 \text{ V}$	<i>voltage from point 2 to point 1</i>
$E_{3-2} = -10 \text{ V}$	<i>voltage from point 3 to point 2</i>
$E_{4-3} = -20 \text{ V}$	<i>voltage from point 4 to point 3</i>
$+ E_{1-4} = -15 \text{ V}$	<i>voltage from point 1 to point 4</i>
0 V	

$$\sum V = 0$$

Kirchhoff's Voltage Law (KVL)

In any closed loop in a circuit, the **sum of all voltages** (rises and drops) equals zero, or in other words, the **algebraic sum of all voltages** in a loop must equal zero.

E.g.,



Loop Order: 3-2-1-4-3

$E_{2-3} = +10 \text{ V}$	<i>voltage from point 2 to point 3</i>
$E_{1-2} = -45 \text{ V}$	<i>voltage from point 1 to point 2</i>
$E_{4-1} = +15 \text{ V}$	<i>voltage from point 4 to point 1</i>
$+ E_{3-4} = +20 \text{ V}$	<i>voltage from point 3 to point 4</i>
0 V	

Exercise 1 (KVL)

Problem: A single loop contains a 12 V battery and two series resistors $R_1 = 2 \Omega$ and $R_2 = 4 \Omega$. Find the loop current I (clockwise).

KVL (following the loop in the direction of current):

$$+12 - I \cdot 2 - I \cdot 4 = 0$$

Answer: $I = 2.000 \text{ A}$ (clockwise).

Exercise 2 (KVL)

Problem: Two meshes share a common resistor R_3 .

- Left mesh: voltage source $V_1 = 10 \text{ V}$, resistor $R_1 = 2 \Omega$, shared $R_3 = 4 \Omega$.
- Right mesh: voltage source $V_2 = 5 \text{ V}$, resistor $R_2 = 3 \Omega$, shared $R_3 = 4 \Omega$.

Assume mesh currents I_1 (left clockwise), I_2 (right clockwise). Find I_1 and I_2 .

KVL for left mesh (clockwise): go around with V_1 rise then drops:

$$V_1 - R_1 I_1 - R_3 (I_1 - I_2) = 0$$

KVL for right mesh (clockwise):

$$V_2 - R_2 I_2 - R_3 (I_2 - I_1) = 0$$

$$\text{Answers: } I_1 = \frac{45}{13} \text{ A} \approx 3.46154 \text{ A},$$

$$I_2 = \frac{35}{13} \text{ A} \approx 2.69231 \text{ A}.$$

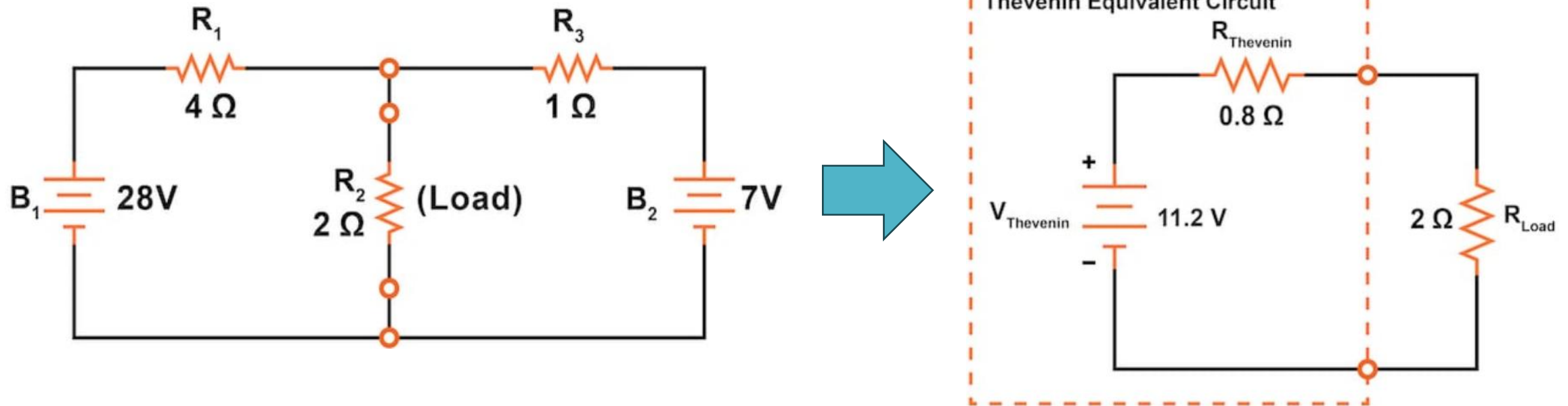
Exercise 3 (Theories Combined)

A 12.0 V battery is connected through a series resistor $R_1 = 4.0 \, \Omega$ to node A. From node A to ground there are two resistors in parallel: $R_2 = 6.0 \, \Omega$ and $R_3 = 12.0 \, \Omega$. In addition, a current source of $I_s = 2.0 \, \text{A}$ injects current into node A (from ground up into node A). Find node voltage V_A , the branch currents I_{R2} , I_{R3} (to ground), and the series current I_{series} coming from the battery into node A.

- Node voltage: $V_A = 10.0 \, \text{V}$.
- Currents to ground: $I_{R2} = \frac{5}{3} \, \text{A} \approx 1.6667 \, \text{A}$, $I_{R3} = \frac{5}{6} \, \text{A} \approx 0.8333 \, \text{A}$.
- Series current from battery: $I_{series} = 0.50 \, \text{A}$.

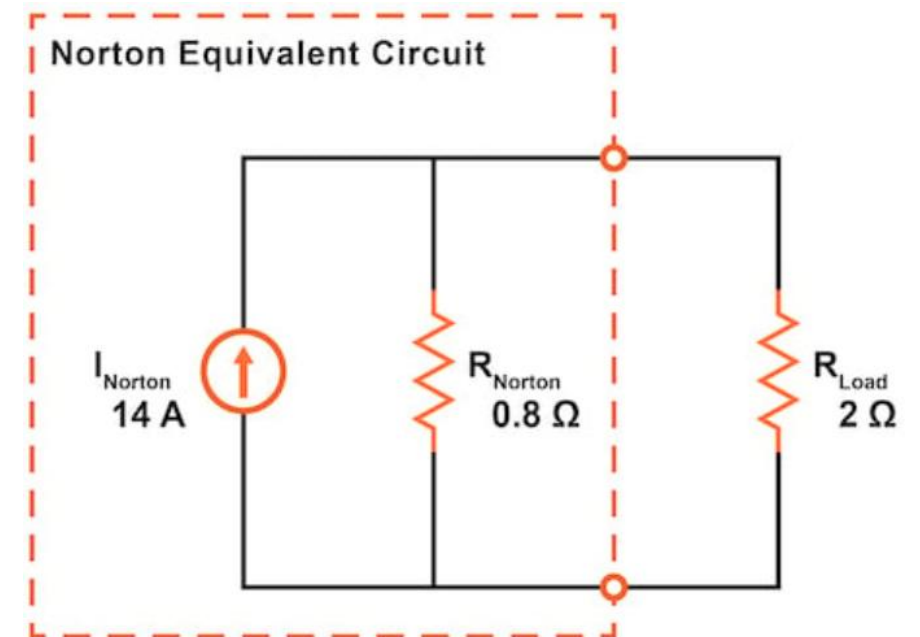
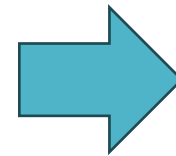
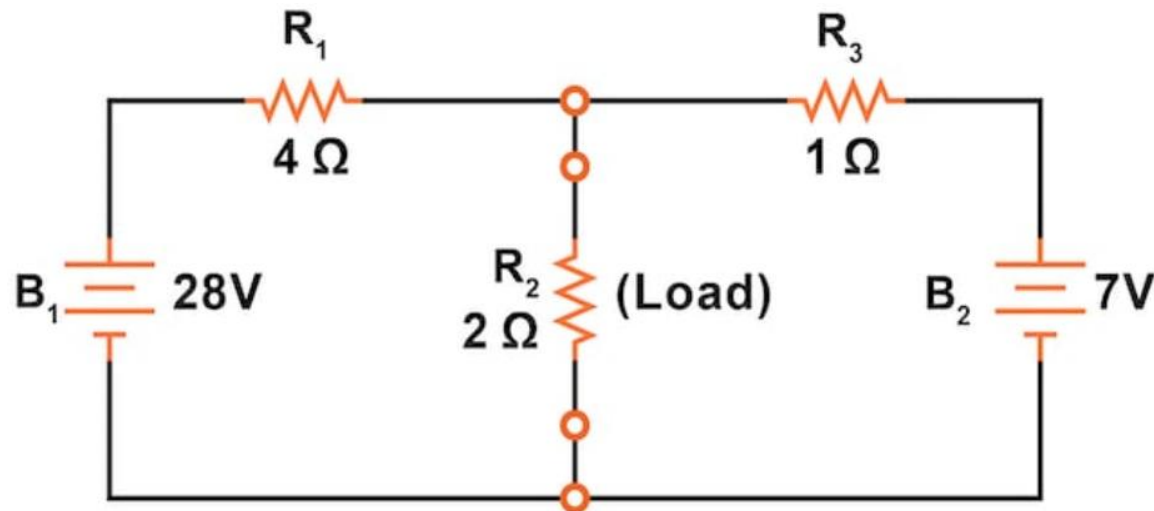
Thevenin's Theorem

Any **linear** two-terminal circuit (not containing any device where $V-I$ is nonlinear such as diodes and transistors) containing voltage sources, current sources, and resistors can be replaced by an **equivalent single voltage source** (V_{th}) in series with a **resistor** (R_{th}).



Norton's Theorem

Any **linear** two-terminal circuit can be replaced by an **equivalent current source** (I_n) in parallel with a **resistor** (R_n).



Exercise (Thevenin & Norton)

Consider this circuit (two terminals are **A** and **B**, where **B = ground**):

A 24 V voltage source $V_s = 24\text{ V}$.

A series resistor $R_1 = 6\ \Omega$ between the source and node A.

A resistor $R_2 = 3\ \Omega$ from node A **to ground** (i.e. from A to B).

The load R_L will be connected between A and ground (B).

Find the **Thevenin** equivalent seen from A–B (open circuit at A).

Find the **Norton** equivalent.

Verify both by connecting a load $R_L = 2\ \Omega$.



- $V_{th} = 8\text{ V}$
- $R_{th} = 2\ \Omega$
- Norton: $I_n = 4\text{ A}$, $R_n = 2\ \Omega$
- With $R_L = 2\ \Omega$: $I_L = 2\text{ A}$, $V_L = 4\text{ V}$